

Repair What Matters: Certified Local Recompute for Persistent 3D Reconstructions

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Persistent 3D reconstructions often outlive their capture session: objects move, AR maps are revisited, or digital twins are edited. Rebuilding every derived representation after each change is safe but can waste work; updating only nearby geometry is cheaper but can leave stale ownership, GPU-publication, visibility, or temporal state that changes the rendered output.

We present *Criticality-Bounded Revision Cones* (CBRC), an incremental recomputation algorithm for selective maintenance of persistent 3D state. A revision cone is the state recomputed after an edit. CBRC closes exact structural dependencies, bounds supported influence left outside the cone at protected outputs, and accepts a local update only when every output bound is within tolerance and its calibrated work is below a measured full-rebuild baseline. Unsupported, supercritical, or uneconomical cases automatically return FULL.

We implement the method in MAVEB on a C++23 reconstruction and Gaussian-rendering stack. Across 1,275 revisions over 85 scenes from 3RScan, ARKitScenes, Bonn RGB-D Dynamic, and trained GraphDECO 3DGS, CBRC selects 935 local repairs and 340 full fallbacks with no observed certificate violation under independent full-after replay; median calibrated selected work is 0.474 of a rebuild. A separate residual-sensitive study on 20 independently selected scenes shows certified non-zero opacity locality in every scene after an opacity-delta refinement, while an unchanged translation control remains at zero. The result is a graphics-systems rule for local recomputation under an explicit output-error contract, not a new reconstruction representation or renderer.

CCS Concepts: • **Computing methodologies** → **Computer graphics**; **Rendering**; *Reconstruction*.

Additional Key Words and Phrases: persistent 3D reconstruction, incremental recomputation, 3D Gaussian Splatting, change-aware rendering, dependency analysis, certified local updates

1 INTRODUCTION

A captured 3D scene is often treated as a finished reconstruction, yet many deployed systems keep that scene alive. Consider a reconstructed room in which one chair is moved. Recomputing the entire representation, every GPU record, and every temporal buffer is safe but wasteful. Recomputing only the chair’s nearby geometry is cheaper, but it can be wrong if the edit also changes ownership, visibility, publication ranges, or retained history. The practical graphics problem is therefore not merely “where did the edit occur?” but “which derived state must change before the updated output can be trusted?”

This paper separates two notions that are easy to conflate. *Computational locality* means that only a subset of the persistent state is recomputed. *Output locality* means that leaving the remaining state untouched changes a protected output—for example an RGB render—by no more than a declared

tolerance. A method can have computational locality without output locality: it may update a small region quickly while leaving stale state that visibly changes the result. Our goal is to admit a local update only when both notions hold under the stated model.

The safe reference is a full rebuild. Classical volumetric and dense reconstruction systems already provide a natural notion of recomputing the complete derived state [7, 8, 19]. Explicit Gaussian and hybrid representations make the potential waste especially visible because large numbers of persistent primitives can remain unchanged after a small edit [13, 14, 17]. Continual Gaussian systems exploit changed regions to focus optimization [2, 29, 30]. We address a different decision: after the edit is known, when may the system deliberately leave some derived state unrecomputed while bounding its effect on the output?

Our answer is *Criticality-Bounded Revision Cones* (CBRC). The name is more complicated than the operation. A revision cone C is simply the set of state blocks that will be recomputed. Starting from the edited blocks, CBRC performs four steps:

- (1) close all dependencies whose correctness is structural and therefore cannot be approximated;
- (2) propagate conservative finite-change bounds through the supported approximate dependencies that remain outside the cone;
- (3) map that unrecomputed influence to each protected output and compare it with the output tolerance; and
- (4) accept the cone only if it is both output-feasible and cheaper than the measured FULL baseline, otherwise enlarge it or return FULL.

This makes fallback part of the algorithm rather than an exceptional failure mode.

The dependency distinction is implementation-facing. A HARD edge represents a relation such as ownership, identity, topology, publication, or unstable temporal invalidation that must remain exact. An ANALYTIC edge is used only where the implementation has a conservative finite-change bound. An EMPIRICAL edge may influence search order, but it cannot certify correctness. Unsupported relations are not guessed; they are promoted to exact closure or force a full rebuild. The graph is constructed from persistent ownership and publication metadata already carried by the engine rather than learned from previous edit outcomes.

For a protected output q , ε_q is the allowed difference from the corresponding full-after result. The planner emits a conservative bound $B_q(C)$ for the state left outside the proposed cone. The central contract is

$$E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q, \quad (1)$$

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where $E_q^{\text{full-ref}}$ is measured independently during evaluation. The left inequality tests whether the certificate was conservative; the right inequality is the planner’s acceptance condition. Separately, the planner compares a calibrated regional-work estimate $W(C)$ with an independently measured W_{full} . The work model does not claim wall-clock speedup; it gives the planner a common scale for heterogeneous counters such as Gaussian updates, GPU publication bytes, and temporal invalidation pixels.

MAVEB is the research implementation of CBRC, and AETHER is the underlying C++23 persistent-3D stack. The system spans reconstruction state, Gaussian primitives, revision transactions, GPU publication, temporal history, and deterministic evidence capture. Exact graph dependencies come from explicit state ownership and back-references. The supported analytic paths include Gaussian display-space influence and stable temporal blending. A dense Python reference planner and an independent full-after Gaussian renderer check the native C++ decision path rather than trusting it by construction. Figure 1 shows the complete local-update decision, and Figure 2 illustrates how an edit expands into a selected recomputation region.

We evaluate two questions separately. First, does the complete system choose useful local updates and fall back as coupling grows? The broad campaign contains 1,275 revisions over 85 scenes and selects 935 LOCAL repairs and 340 FULL fallbacks, with no observed certificate violation under independent replay and a median calibrated selected-work ratio of 0.474. Second, can the method certify deliberately non-zero residuals rather than only exact selected-to-full equality? A separate residual-sensitive sequence exposes a conservative certificate failure, changes only the opacity residual bound, and confirms the revised mechanism on 20 independently selected scenes while keeping translation as an unchanged control.

The scope of the claim is intentionally narrower than universal “safe editing.” CBRC certifies a local recomputation only under the dependencies, renderer assumptions, quantities of interest, tolerances, and work calibration that are explicitly represented. If those assumptions are unsupported or too conservative, the method returns FULL. The paper therefore evaluates a graphics-systems algorithm for reducing recomputation, not a new reconstruction model, an edit detector, or a general proof of scene correctness.

The technical contributions are:

- a cross-derived-state local-recompute formulation for persistent 3D reconstructions that distinguishes exact structural dependencies from conservatively bounded output influence;
- an executable acceptance rule that combines dependency closure, output-specific error bounds, a measured work crossover, and automatic full fallback in one revision algorithm;

- implementation-backed display-space bounds for Gaussian revisions, including an opacity-delta refinement that can certify bounded non-zero residuals under explicit assumptions; and
- an end-to-end implementation and evaluation across reconstruction, Gaussian, GPU-publication, and temporal state, with independent full-after replay, 85-scene broad testing, and a separate 20-scene residual-sensitive confirmation.

2 RELATED WORK

Persistent reconstruction and map maintenance. Volumetric integration gave reconstruction systems a durable way to accumulate range observations [7], and surfels offered an explicit point-oriented alternative [22]. KinectFusion later showed real-time dense fusion and tracking [19]; ElasticFusion maintained and corrected dense surfel maps without a conventional pose graph [25]; and BundleFusion reintegrated surfaces online while maintaining global consistency [8]. All of these systems make map state persistent enough that revision becomes meaningful, although their update rules are closely tied to the reconstruction or consistency mechanism itself. CBRC looks at the revision one level above that: which derived state really has to be recomputed, what may remain outside it, and at what point should the incremental attempt be abandoned.

Neural and Gaussian mapping. NeRF established continuous neural radiance fields for novel-view synthesis [18], while iMAP and NICE-SLAM brought learned implicit representations into online mapping and SLAM [23, 31]. 3D Gaussian Splatting (3DGS) moved back toward an explicit anisotropic representation while retaining real-time rendering [14]; SplatTAM and Gaussian Splatting SLAM then adapted Gaussian state to online RGB-D and monocular mapping [13, 17]. Dynamic 3D Gaussians and 4D-GS extend the representation over time [16, 27]. These representations make revision especially tangible because individual persistent primitives and their support are exposed. Still, the CBRC formulation itself is representation-agnostic.

Continual and evolving Gaussian scenes. CL-Splats separates changed and static scene components so that Gaussian optimization can stay focused while older scene states are preserved [2]. GaussianUpdate uses change-specific update strategies and visibility-aware replay for continual radiance fields [30]. GaME addresses long-term evolving RGB-D scenes by adapting the Gaussian map while managing stale observations [29], and EliGSiR studies continual Gaussian mapping under a bounded compute budget through adaptive view scheduling, supervision fidelity, and geometry growth [9]. Local updating of changing Gaussian scenes is therefore established prior art. The distinction in CBRC is the contract

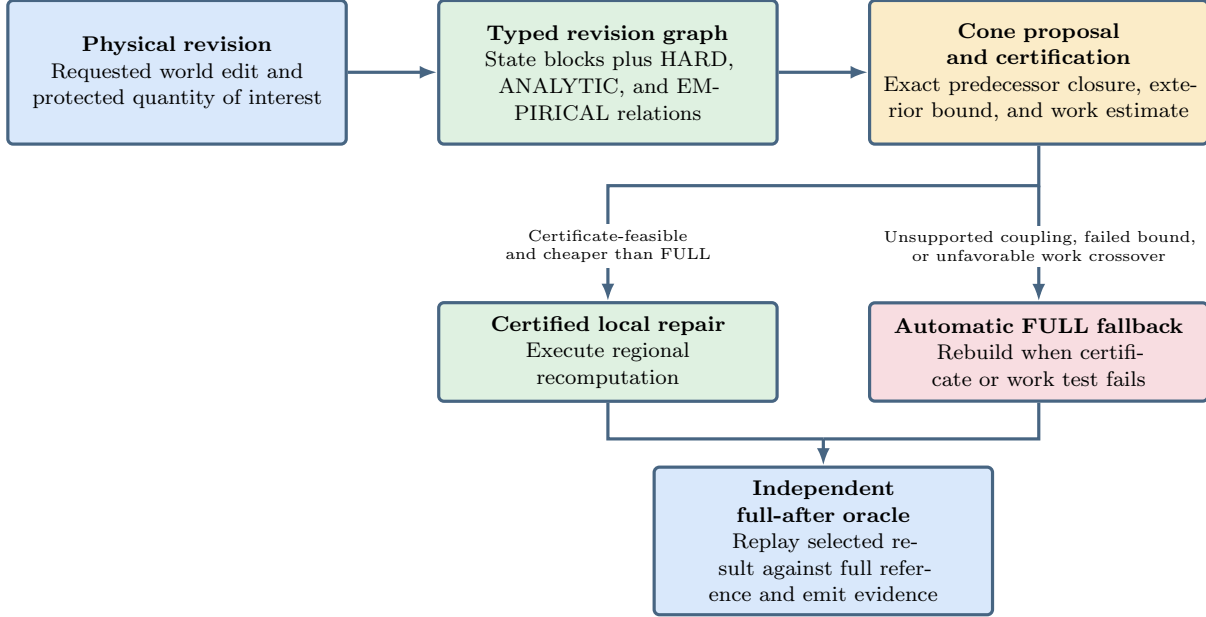


Fig. 1. CBRC local-recompute decision and independent verification path. A physical edit enters a typed dependency graph. Exact HARD predecessors are closed, conservative ANALYTIC influence is propagated through the state left unrecomputed, and output bounds are checked against tolerance. A local repair is executed only when it is certified and cheaper than the full baseline; otherwise the system falls back to FULL. The selected output is then checked against an independently rendered full-after reference.

it enforces: cross-representation exact dependencies and conservative output bounds are checked together, locality is admitted under Eq. 1, and FULL remains an explicit outcome when certification or the work comparison fails.

Editing, change detection, and identity. GaussianEditor and SC-GS demonstrate precise, controllable local editing of Gaussian representations [6, 12]. Primitive-space change detection further suggests that Gaussian attributes themselves can reveal structural and appearance changes [10], while Consistent Instance Field studies persistent probabilistic identity in dynamic Gaussian scenes [28]. These works set a clear boundary for our contribution. Local editing, primitive-level change detection, and persistent identity are upstream capabilities; CBRC begins after a revision has been specified or detected and asks which derived state may safely remain unrepaired for a protected output.

Incremental rendering, change propagation, and output-directed error control. Dependency-aware recomputation has a long history in graphics. Bala et al. connect edited world regions to bounded-error radiance interpolants with ray segment trees [3], and Wörster et al. derive a dependency graph from scene-graph changes to rendering caches so those caches can be updated incrementally [26]. Beyond graphics, self-adjusting computation formalizes correct change propagation under mutation and reuse [1]; goal-oriented error estimation directs numerical work toward selected outputs

instead of a global state norm [20]. These lines establish the ingredients that make local recomputation possible.

The gap addressed here is the decision boundary inside a modern persistent 3D pipeline. Reconstruction state, explicit Gaussian primitives, GPU publication, and temporal history do not share one update unit or one error metric. CBRC turns those heterogeneous states into a single executable local-recompute decision: exact relations are closed, bounded relations contribute to an output-error certificate, heterogeneous work is compared with a full baseline, and unsupported or uneconomical cases return FULL. The underlying graph and norm tools are standard; the technical contribution is their realization as a cross-derived-state graphics algorithm with an independent full-after check.

3 PROBLEM FORMULATION

3.1 Persistent 3D state and revision cones

Let the persistent system state be partitioned into blocks

$$V = \{1, \dots, n\}. \quad (2)$$

A block is a unit that can be recomputed or retained independently by the implementation: an observation region, TSDF block, mesh patch, texture page, material record, Gaussian subset, GPU publication range, temporal-history tile, or another derived-state unit. The partition is computational rather than purely spatial.

An edit provides a direct change envelope $b \in \mathbb{R}_{\geq 0}^n$ for the blocks whose values are known to change. A *revision cone*

How Local Gaussian Revisions Propagate

From intended physical edit to certified local repair or FULL fallback.

MAVEB / CBRC
Certified Local Revision of
Persistent Gaussian Worlds

Edit locally.
Prove the residual.
Fall back to FULL
when necessary.

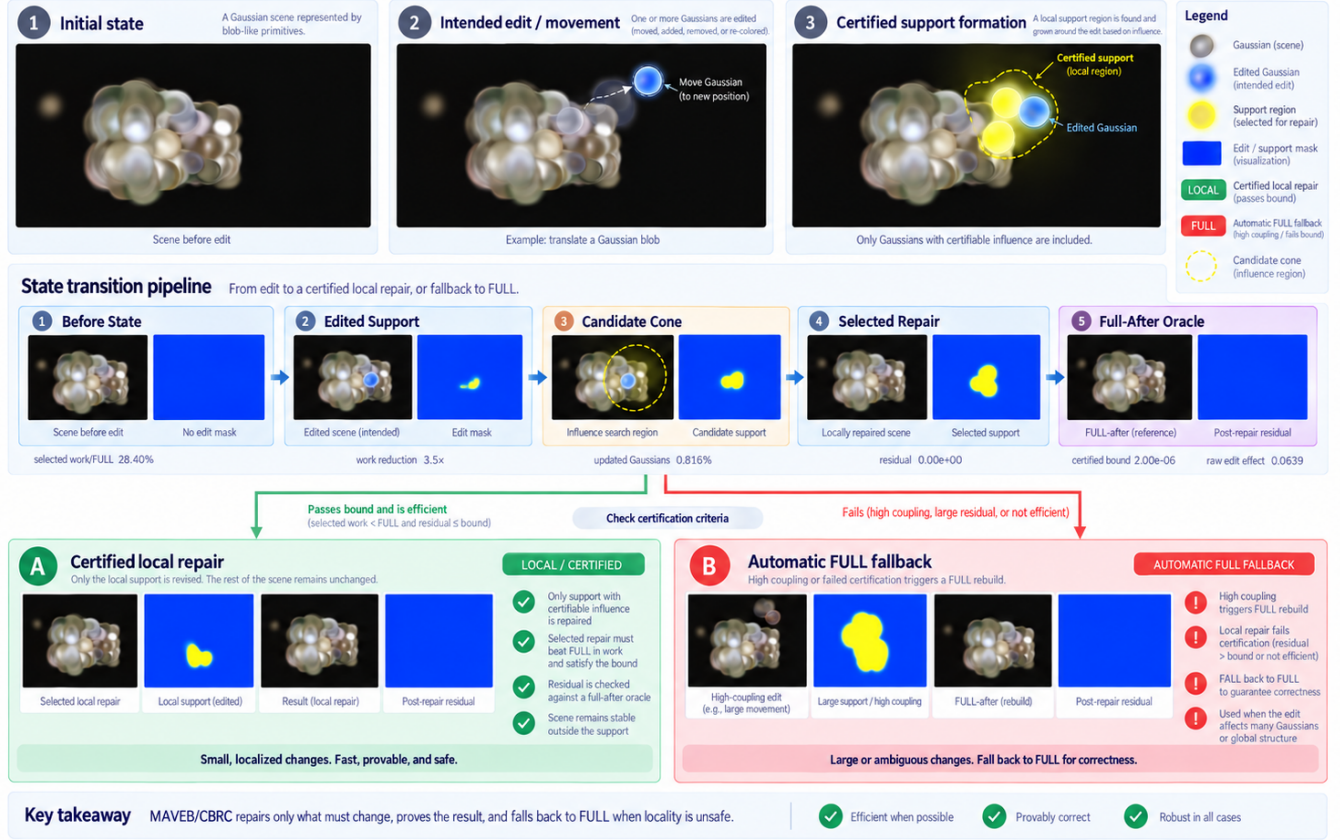


Fig. 2. From a physical edit to a selected recomputation region. The intended Gaussian change defines the initial affected state. CBRC expands that state until exact dependencies are closed and the influence left outside the region is within the declared output tolerance. The local result is used only when its estimated regional work is also below the full baseline; otherwise the algorithm selects a full rebuild. Blue marks the intended edit and yellow the state selected for recomputation. The supplied image is reproduced unchanged. Its embedded 28.40% work ratio, 3.5× work reduction, and other micro-metrics are illustrative values from the graphic and are separate from the aggregate campaign estimates reported in Section 7.

$C \subseteq V$ is the set of blocks selected for recomputation. The exterior

$$O = V \setminus C \quad (3)$$

is intentionally retained from the previous state. Choosing a small C gives computational locality; proving that the retained exterior changes the protected output by at most the allowed tolerance gives output locality. CBRC requires both.

The planner does not need to execute the current revision globally in order to decide whether to try a local update. The initial change envelope comes from the requested edit and the current persistent metadata. Analytic bounds use the known

before/after parameters of edited state and implementation-defined dependency bounds. Only the evaluation oracle performs an independent full-after replay, and it is used to test the planner rather than to make the planner's decision.

3.2 Where dependencies come from

Each dependency has one of three semantics (Table 1). The categories are assigned by the implementation, not learned from historical success/failure labels. Exact ownership, identity, topology, publication, and invalidation relations become HARD edges. A relation becomes ANALYTIC only when the implementation can produce a conservative finite-change upper bound for it. Measurements or learned signals may be stored as EMPIRICAL edges to improve search order, but they cannot make a candidate certifiable.

Table 1. Revision-edge semantics in CBRC.

Type	Meaning	Certificate role
HARD	Exact ownership, identity, topology, publication, or invalidation dependency	Must be closed exactly
ANALYTIC	Conservative implementation-backed finite-change upper bound	May certify exterior influence
EMPIRICAL	Measured or learned influence used for ordering or scheduling	Never treated as proof

We keep this separation conservative on purpose. Dependencies from observations to TSDF state, TSDF to mesh ownership, meshes to texture-page identity, and Gaussian records to GPU publication remain exact in v1. Unstable temporal validation is treated the same way. Gaussian-to-image influence and stable temporal blending have implemented ANALYTIC bounds. An EMPIRICAL-only dependency without a conservative analytic bound is promoted to exact predecessor closure for certification; its measured gain may still guide scheduling, but never serves as a soft safety bound. Relations lacking a conservative argument stay exact even when a softer treatment would shrink the cone.

3.3 Output and work contracts

A protected quantity of interest q is an output whose deviation from a full rebuild matters to the application. In this paper the principal example is rendered RGB output on declared pixels. Let $\varepsilon_q \geq 0$ be the maximum allowed deviation. For a candidate cone C , the planner computes $B_q(C)$, a conservative bound on the effect of the state left unrecomputed. The candidate is output-feasible only when

$$B_q(C) \leq \varepsilon_q \quad (4)$$

for every protected output. During evaluation, Eq. 1 additionally checks this certificate against an independently produced full-after reference.

Local recomputation must also be cheaper than rebuilding. Native work counters have different units, so they are not added directly. If $n_d(C)$ is the amount of native work in domain d and κ_d is a cost per native unit measured in a separate calibration run, the planner uses

$$W(C) = \sum_d \kappa_d n_d(C), \quad (5)$$

and compares it with an independently measured W_{full} . The coefficients κ_d are frozen before the reported campaign; they are not fitted after seeing whether an individual revision succeeds. This is a heterogeneous work proxy, not an end-to-end timing claim.

4 ALGORITHM: CERTIFIED LOCAL RECOMPUTE

4.1 Exact predecessor closure

If a repaired state block v requires an exact predecessor u , then the predecessor must also be inside the cone:

$$v \in C, u \in \text{Pred}_{\text{HARD}}(v) \implies u \in C. \quad (6)$$

This condition is checked before any approximate reasoning. If Eq. 6 is violated, the candidate is invalid regardless of how small a local sensitivity estimate happens to be.

4.2 Conservative exterior propagation

For ANALYTIC dependencies we use a componentwise non-negative influence matrix $K \in \mathbb{R}_{\geq 0}^{n \times n}$. Each entry K_{vu} upper-bounds the normalized finite change that block u can transmit to block v . Let z_C collect known change bounds for the repaired state. The unrepaired exterior then satisfies

$$\delta_O \leq b_O + K_{OC} z_C + K_{OO} \delta_O. \quad (7)$$

When the exterior response is certifiable and $\rho(K_{OO}) < 1$, the classical resolvent gives

$$G_O = (I - K_{OO})^{-1} = I + K_{OO} + K_{OO}^2 + \dots, \quad (8)$$

and therefore

$$\hat{\delta}_O = G_O (b_O + K_{OC} z_C), \quad \delta_O \leq \hat{\delta}_O. \quad (9)$$

The native v1 planner uses sparse finite propagation on supported DAG exteriors. An analytic cycle outside that supported case triggers fail-closed behavior: the cone expands until the unsupported exterior disappears, or the result becomes FULL. Equation 9 is the reference formulation; production support is limited to the stated DAG case.

4.3 Quantity-of-interest certificate

Let R_q map state perturbation to the protected output quantity. The exterior contribution is bounded by

$$B_q(C) = \left\| |R_{q,O}| \hat{\delta}_O \right\|_{\infty}. \quad (10)$$

A candidate is certificate-feasible only when $B_q(C) \leq \varepsilon_q$ for every declared output. As a result, the same geometric change can lead to different revision decisions when the protected outputs differ. This is intentional: the certificate is tied to the protected output, avoiding dependence on an internal state norm whose relation to visible error may vary.

4.4 Normalized certificate ratio

For search and diagnosis we summarize all output constraints with the worst normalized certified residual. We call this scalar the *revision criticality* of a candidate cone

$$\begin{aligned} \chi(C) &= \max_q r_q(C), \\ r_q(C) &= \begin{cases} B_q(C)/\varepsilon_q, & \varepsilon_q > 0, \\ 0, & \varepsilon_q = B_q(C) = 0, \\ +\infty, & \varepsilon_q = 0 < B_q(C). \end{cases} \end{aligned} \quad (11)$$

The decision boundary is $\chi(C) = 1$. Values at or below one are certificate-feasible; a value above one means that at least

one protected output is supercritical and the cone must grow or fall back to FULL. We use the phrase *criticality-bounded* in this narrow sense of a normalized output-error boundary. It is not meant to suggest a new model of physical critical phenomena.

4.5 Gaussian display-space bound

For a projected Gaussian i at pixel p , MAVEB starts from the projected opacity model

$$\alpha_i(p) = o_i \exp\left(-\frac{1}{2}q_i(p)\right), \quad (12)$$

where o_i is peak opacity and $q_i(p)$ is the squared Mahalanobis distance in projected Gaussian space. In the bound that follows, α_i means the effective value after the renderer’s clamp and cutoff rules, not simply the raw value from Eq. 12. For an edited set E , we define the aggregate opacity mass

$$A_p(E) = 1 - \prod_{i \in E} (1 - \alpha_i(p)). \quad (13)$$

Proposition 1 (Gaussian finite-edit display bound). Let U denote the unchanged Gaussian state and E_0, E_1 the complete before/after edited sets at a protected pixel p . Under the assumptions below,

$$\begin{aligned} \Delta_p &= \|R_p(U \cup E_0) - R_p(U \cup E_1)\|_\infty, \\ \Delta_p &\leq C_{\text{color}} \min(1, A_p(E_0) + A_p(E_1)). \end{aligned} \quad (14)$$

Assumptions. The proposition rests on four conditions. First, compositing is front-to-back, with effective post-clamp/post-cutoff alpha in $[0, 1]$. Second, each protected color channel, including the background and the renderer’s bounded spherical-harmonic evaluation, lies in an interval of width C_{color} . Third, E_0 and E_1 include every Gaussian whose projected opacity, color, insertion/deletion state, or depth relation to unchanged support can change because of the edit. Finally, renderer behavior that is not captured by these bounded quantities is treated as unsupported and is promoted to an exact dependency or a FULL fallback.

Proof sketch. For a single edited set, front-to-back compositing can alter a channel by no more than the channel range times the opacity mass through which edited support replaces, attenuates, or reveals unchanged radiance. This gives $C_{\text{color}} A_p(E)$. Insertion and deletion are the special cases in which one edited set is empty. For a general before/after edit, we pass through the common unchanged state U and apply the triangle inequality, obtaining $C_{\text{color}}(A_p(E_0) + A_p(E_1))$. The result is then capped by the largest possible channel-range change, C_{color} . The image-level certificate takes the maximum of Eq. 14 over the declared protected pixels. The supplement gives the algebra in more detail.

Opacity-only delta refinement. Equation 14 is intentionally general, but it is insensitive to how small an opacity-only edit is: the before/after opacity masses can remain large even when their difference is tiny. For paired Gaussian records whose position, projected covariance, depth order, rotation,

and spherical-harmonic color are unchanged, the implementation uses the compositor’s coordinate-wise Lipschitz property in effective alpha. Let $\alpha_i^0(p)$ and $\alpha_i^1(p)$ be the renderer-admitted before/after alpha values. On pixels where at least one admitted alpha changes,

$$\Delta_p \leq C_{\text{color}} \min\left(1, \sum_i |\alpha_i^0(p) - \alpha_i^1(p)| + 0.002\right). \quad (15)$$

The 0.002 term covers the reference compositor’s two-sided early-termination discrepancy: each endpoint may omit at most 0.001 transmittance after accumulated alpha exceeds 0.999. The allowance is added only where an opacity delta is actually admitted, so pixels outside changed support retain a zero bound. The routine fails closed if any paired primitive changes geometry, rotation, covariance, spherical-harmonic coefficients, or camera visibility unexpectedly. V5 and v6 use this refinement for opacity residuals; their translation control continues to use Eq. 14.

4.6 Stable temporal history

For stable temporal validation, let E_c denote current-frame error, E_h retained-history error, E_n neighborhood or clamping error, and $w \in [0, 1]$ the history weight. With $E_r = \max(E_h, E_n)$,

$$E_{\text{resolved}} \leq (1 - w)E_c + wE_r. \quad (16)$$

If validation, reprojection, or disocclusion can itself change, MAVEB does not apply Eq. 16; it promotes the relation to HARD invalidation. The temporal case is a useful example of the design rule we follow throughout: a soft bound is used only for as long as its own preconditions can be certified.

4.7 Cone selection and automatic full fallback

The v1 search still needs a numerical ordering when a declared tolerance is zero. For stable exteriors it uses $\tilde{\chi}(C) = \max_q B_q(C) / \max(\varepsilon_q, 10^{-15})$ only as a search score. The floor does not relax acceptance, which still uses $B_q(C) \leq \varepsilon_q$. For an expansion $C \rightarrow C'$ with positive added work, the utility is

$$\text{utility}(C \rightarrow C') = \frac{\tilde{\chi}(C) - \tilde{\chi}(C')}{W(C') - W(C)}. \quad (17)$$

Each candidate is closed over its exact predecessors before it is certified again. The search continues until it finds a cone that is both certificate-feasible and cheaper than FULL; if no such cone is reached, the planner returns the full fallback. This procedure returns a certified feasible cone; global minimum-work optimality is outside its guarantee.

5 SYSTEM IMPLEMENTATION

MAVEB is the experimental implementation of CBRC and AETHER is the underlying C++23 persistent-3D software stack. The native path uses a sparse typed revision graph, sparse analytic propagation over supported DAG exteriors, exact predecessor closure, work-aware cone search, and deterministic certificate serialization. A separate dense Python implementation follows the mathematical contract directly

and is used as a reference check. The native and Python planners evaluate the same campaign inputs and their final decisions are compared rather than assumed equivalent.

Dependency construction. The graph is built from explicit state relations already present in the engine. Observation identifiers map to reconstruction blocks; reconstruction blocks own mesh regions; persistent Gaussian entities map to GPU publication ranges; temporal tiles record the screen-space history they retain. These relations become exact edges when stale state could invalidate identity, ownership, topology, publication, or temporal validity. Analytic edges are enabled only for implemented bounded mechanisms such as Gaussian display-space influence and stable temporal blending. There is no learned dependency discovery in the reported planner, and previous edit outcomes are not used as correctness evidence.

Revision transactions. The production path records persistent Gaussian entity translations, immutable before/after sidecars, exact camera binding, a production image certificate, and the planner’s temporal repair or retention decision. In headless operation, the system loads a persistent world, applies indexed Gaussian transactions, writes the new durable revision and sidecars, evaluates the Gaussian certificate, runs the output-cone planner, and emits canonical evidence JSON. If the consistency checks do not pass, the revision fails closed instead of publishing a partly updated world.

Cross-derived-state accounting. A `LocalityLedger` tracks observation, TSDF, mesh, texture-page, material, Gaussian, GPU-publication, and temporal work; ledger/mesher disagreement is an error. Counts retain their native units until a versioned calibration is applied. The frozen public campaign calibrates only four measured domains: Gaussian inspection, Gaussian update, GPU publication bytes, and temporal invalidation pixels.

Independent oracle. The Gaussian oracle is independent of the planner. It renders before and full-after states, then forms a support replay from full-after values on the selected support and before-state values outside it. The per-pixel exterior residual is checked against Eq. 1; certificate disagreement or a residual above the emitted bound is a correctness failure. Evidence bundles record hashes, workflow IDs, and native/reference output-cone parity.

Representation scope. The broad campaign contains two representation classes. Seventy-two scenes from 3RScan [24], ARKitScenes [5], and the Bonn RGB-D Dynamic Dataset [21] use dataset-native geometry seeded into deterministic Gaussian fields after the public capture assets are normalized into the benchmark coordinate convention. Thirteen scenes use pinned pretrained GraphDECO 3DGS representations from the standard evaluation suite: nine Mip-NeRF 360 scenes [4], Train and Truck from Tanks & Temples [15], and Dr Johnson and Playroom from Deep Blending [11]; the

Gaussian representations preserve spherical-harmonic coefficients, opacity, anisotropic scale, and rotation [14]. Ownership is deterministic and spatial in both paths and serves revision bookkeeping; semantic segmentation is outside its role. Selected-to-FULL render fidelity is measured within the evaluated representation, separately from reconstruction quality against original camera photographs.

6 EVALUATION

6.1 Questions and protocol

We evaluate CBRC as an incremental graphics-systems algorithm rather than as a reconstruction-quality method. Four questions drive the experiments. First, when the planner publishes a LOCAL result, does independent full-after replay stay within the emitted bound and declared tolerance? Second, how does the opportunity for locality change with coupling strength and edit type? Third, how much calibrated work does the selected decision require relative to a full rebuild? Fourth, do those behaviors persist across dataset families, seeded and trained Gaussian representations, and alternative selection policies?

The broad campaign is fixed before outcome inspection. It contains 1,275 revisions over 85 scenes: 600 cases on 40 3RScan reference/rescan worlds [24], 300 cases on 20 ARKitScenes worlds [5], 180 cases on 12 Bonn dynamic RGB-D sequences [21], and 195 cases on 13 pinned pretrained GraphDECO 3DGS scenes drawn from Mip-NeRF 360, Tanks & Temples, and Deep Blending [4, 11, 14, 15]. Every scene contributes the same 15-case coupling template. The evaluated edit families are translation, rotation, uniform-scale, and opacity. Fallbacks remain part of the evaluation because returning FULL is an explicit algorithmic outcome rather than a discarded case.

Each case records the source edit, chosen cone, decision (LOCAL or FULL), certificate, independent full-after replay, work, baselines, ablations, and provenance. The visual audit compares the selected output with the independent FULL-after render at the same camera.

Work metric. The headline work ratio uses the frozen heterogeneous work model from Eq. 5. The four calibrated domains are Gaussian inspection, Gaussian update, GPU publication bytes, and temporal invalidation pixels. Bootstrap intervals use 5,000 resamples. The work ratio is an execution-independent accounting metric; paired local-versus-full wall-clock timing remains outside the current claim.

Statistical reporting. Local and violation rates in the broad campaign use Wilson 95% intervals. Median work ratios use bootstrap 95% intervals. Paired baseline comparisons count cases in which both policies certify, then report which policy uses less calibrated work, ties, and an exact two-sided sign test when the comparison is informative. The confirmatory residual study uses the scene as its primary sampling unit. Its 95% intervals resample scenes within each fixed dataset stratum, preserving the five-scene-per-dataset design in every

bootstrap replicate; the 3,840 parameter cases are pooled evidence, not 3,840 independent samples.

6.2 Residual-sensitive protocol

The non-zero-residual experiments are frozen separately from the 1,275-case base campaign. The v4 diagnostic uses four scenes, four locality profiles, exact and 1/1024 residuals, and an epsilon ladder of 1, 4, 16, 32 divided by 255. It is retained unchanged after inspection. V5 changes the image certificate only for opacity residuals: corresponding Gaussian records must be identical except for opacity logit, and the bound uses the admitted change in pixel alpha rather than the older magnitude-insensitive opacity envelope. Translation remains on the older envelope as a control. V5 uses four scenes, residuals exact, 1/4096, 1/1024, and 1/256, and six epsilon values from 1/255 to 32/255.

V6 is a confirmatory breadth run of the unchanged v5 mechanism. Thresholds, residual scales, edit magnitudes, locality profiles, and the control family remain fixed. Five deterministically ranked scenes are selected from each of 3RScan, ARKitScenes, Bonn RGB-D Dynamic, and trained GraphDECO 3DGS, giving 20 scenes and 3,840 pooled parameter cases. Confirmation gates are fixed before execution, and outcome inspection cannot trigger scene or case replacement.

6.3 Baselines and ablations

The method-level suite includes FULL, exact-only repair, fixed graph radii, changed-fraction selection, empirical ordering followed by analytic re-certification, and CBRC. Structural ablations remove individual mechanisms while retaining the frozen campaign. We preserve neutral ablations as neutral results. A mechanism can remain important to the contract even when this real matrix does not separate its aggregate work from the full method; deterministic mechanism-isolation tests are retained separately for falsifiability.

7 RESULTS

7.1 Overall broad campaign

All 1,275 frozen cases complete across all 85 scenes. CBRC selects 935 certified LOCAL repairs and 340 FULL fallbacks, giving a local rate of 73.33% (95% CI 70.84–75.69%). We observe zero certificate violations; the corresponding 95% Wilson interval for the violation rate is 0–0.3004%. Both local and fallback outcomes appear in every representation class.

The median calibrated selected work ratio is

$$\text{median } W(C)/W_{\text{full}} = 0.4744733, \quad (18)$$

with bootstrap 95% CI [0.46657, 0.47885]. At the median, this is 52.55% lower calibrated work than FULL. The minimum observed ratio is 0.07030; fallback cases have ratio 1 by construction.

The identical 73.33% local rate across datasets is a consequence of applying the same frozen 15-case coupling template

to every scene; it should not be interpreted as evidence that the datasets behave identically. Dataset-dependent work differs substantially: the median ratio is 0.341 on 3RScan and 0.519 on trained 3DGS.

7.2 Coupling controls fallback behavior

The fail-closed envelope becomes visible when the campaign is grouped by coupling regime. All low and medium cases remain local. High coupling produces 170 local repairs and 85 fallbacks, while adversarial coupling produces 85 local repairs and 255 fallbacks. The frozen matrix therefore shows a monotonic shift toward fallback as coupling strengthens.

7.3 Generalization across edit families

The same revision contract is exercised across four edit families. Opacity remains local in all 255 cases. Translation and rotation each remain local in 255 of 340 cases. Uniform scale is the hardest family in the frozen matrix, with 170 local repairs and 170 fallbacks. No edit family contains an observed certificate violation.

7.4 Independent selected-render audit

The selected render is byte-identical to the independent FULL-after render in all 1,275 cases on the evaluated benchmark views: the selected exact-case rate is 1.0. For fallback cases this equality follows from selecting FULL. For the 935 LOCAL cases it means the executed repair reproduces the full-after raster exactly at 8-bit RGB precision on the audited view.

This result is strong evidence for the selection/fallback implementation, but it is deliberately separated from certificate tightness. Because the measured selected-to-FULL residual is zero throughout this broad campaign, the matrix does not demonstrate a useful case with

$$0 < E_q^{\text{full-ref}} \leq B_q(C) \leq \varepsilon_q. \quad (19)$$

The separately frozen residual-sensitive sequence below targets exactly this regime without changing the broad campaign.

7.5 From diagnostic failure to confirmatory non-zero locality

The first residual-sensitive freeze, v4, is a negative result. Across 512 cases, the 256 exact-residual cases remain local, while every one of the 256 non-zero-residual cases falls back to FULL. There are no certified non-zero local cases and no tolerance crossovers. Postmortem analysis attributes all 256 non-zero fallbacks to the post-repair certified bound exceeding epsilon. The outcome is insensitive to the locality profile and to the tested epsilon ladder. This rules out a convenient interpretation in which the test scenes were simply too globally coupled: the bottleneck is the certificate used for the residual.

V5 changes one mechanism. For opacity-only residuals, geometry, covariance, rotation, depth, and spherical-harmonic

Table 2. Results by dataset/representation. Work is calibrated selected work divided by the frozen FULL baseline. Confidence intervals are Wilson intervals for local rate and bootstrap intervals for median work.

Dataset / representation	Cases	LOCAL	FULL	Local rate (95% CI)	Median work/FULL (95% CI)
3RScan	600	440	160	0.733 [0.697, 0.767]	0.341 [0.337, 0.341]
ARKitScenes	300	220	80	0.733 [0.681, 0.780]	0.483 [0.480, 0.494]
Bonn RGB-D Dynamic	180	132	48	0.733 [0.664, 0.793]	0.475 [0.474, 0.488]
Trained GraphDECO 3DGS	195	143	52	0.733 [0.667, 0.790]	0.519 [0.513, 0.541]
Overall	1275	935	340	0.733 [0.708, 0.757]	0.474 [0.467, 0.479]

Table 3. Decision envelope by coupling regime.

Regime	Cases	LOCAL	FULL	Median work/FULL
Low	340	340	0	0.415
Medium	340	340	0	0.445
High	255	170	85	0.510
Adversarial	340	85	255	1.000

Table 4. Outcomes by edit family.

Edit	Cases	LOCAL	FULL	Median work/FULL
Opacity	255	255	0	0.457
Rotation	340	255	85	0.453
Translation	340	255	85	0.475
Uniform scale	340	170	170	0.770

color remain fixed, while the certificate accumulates the actual change in admitted per-pixel alpha with a conservative transmittance allowance. Translation continues to use the old opacity-envelope certificate. All certificate-routing, monotonicity, selected-output, and repair-certificate validity gates pass. Of 384 opacity cases, 251 select LOCAL and 133 select FULL; 142 are certified non-zero local cases and 29 stress keys cross from full to local as epsilon changes. The 384 translation controls contain 96 exact local cases and 288 full fallbacks, with zero non-zero local cases and zero crossovers. With translation held as the control, this separation attributes the restored locality to certificate sensitivity.

V6 repeats the same mechanism across 20 independent scenes. The confirmatory audit passes every predeclared gate, with no selected-certificate or repair-certificate violation. Non-zero opacity locality occurs in all 20 scenes and all four dataset families; opacity crossover also occurs in all 20 scenes. Across scenes, the mean opacity non-zero-local rate is 0.477, with a dataset-stratified scene-bootstrap 95% interval of [0.432, 0.517]. Translation remains at 0.000 [0.000, 0.000], so the paired opacity-minus-translation difference is numerically the same 0.477 [0.432, 0.517]. The generic audit contains 687 certified non-zero local cases and 151 tolerance-crossover groups.

The same-opacity legacy-envelope counterfactual provides a direct mechanism check in v6. Holding the physical opacity edit fixed, the delta-sensitive certificate rescues a mean 0.531 of cases per scene that the legacy envelope would reject (95% interval [0.512, 0.549]); all 20 scenes contain at

least one such rescue. The median scene-level ratio of legacy residual bound to delta-sensitive bound is 122.3. This comparison addresses sensitivity; tightness remains a separate question. The accepted cases remain far from the tolerance boundary: the largest scene maximum of measured residual divided by epsilon is 0.0812, the median of those scene maxima is 0.0195, and there are no near-boundary local cases. The median scene-level residual effectivity is approximately 1168, so conservatism remains measured alongside the positive locality result.

7.6 Baselines and ablations

The paired statistics expose both useful separation and neutral mechanisms. Against FULL, all 1,275 cases are jointly certified; CBRC uses lower calibrated work in 935 and ties in 340, with exact two-sided sign-test $p = 6.89 \times 10^{-282}$. Replacing the exterior tail with a global-norm bound produces the same 935-versus-340 separation, showing that the localized tail is the dominant source of work reduction on this matrix.

Several comparisons are neutral. Empirical ordering followed by analytic re-certification ties CBRC in all 1,275 jointly certified cases. Five ablations also tie on aggregate work: Gaussian bound, hard/soft distinction, predecessor closure, QoI specialization, and temporal bound. Exact-only and fixed-radius policies certify 340 of 1,275 cases. The changed-fraction policy certifies 537. Removing fallback leaves 340 certified cases. We retain these results because the experiment characterizes the contract instead of forcing every mechanism to separate.

8 DISCUSSION

8.1 What the method adds

The broad campaign shows where the algorithm is useful and where it deliberately gives up locality. A local update is not accepted because the edit is spatially small; it is accepted only when exact dependencies are closed, the retained exterior is bounded at the protected output, and the estimated work remains below the full baseline. In the 1,275-case campaign that opportunity exists in 935 cases (73.33%) and disappears in 340 cases (26.67%), with fallback becoming more common as coupling strengthens. The method therefore has a measurable operating region rather than assuming that most real edits are local.

The technical contribution is the decision rule across heterogeneous persistent graphics state. Existing reconstruction,

Table 5. Residual-sensitive evidence sequence. Pooled case counts describe frozen parameter grids; v6 inference uses 20 scenes as the primary sampling units, stratified five per dataset. Non-zero LOCAL means a selected local repair with a measured non-zero residual that remains within its certified bound and tolerance.

Stage	Frozen population	Non-zero LOCAL evidence	Main diagnostic / control result
v4 diagnostic	4 scenes, 512 cases	0 cases; 0 tolerance crossovers	All 256 non-zero-residual cases fall back because the certified post-repair bound exceeds epsilon.
v5 mechanism	4 scenes, 768 cases	142 opacity cases across all 4 datasets; 29 crossover keys	Opacity: 251 LOCAL/133 FULL. Translation control: 0 non-zero LOCAL, 0 crossovers.
v6 confirmation	20 scenes, 3,840 cases	687 cases; all 20 scenes and all 4 datasets; 151 crossover groups	Scene-level opacity-translation difference 0.477, 95% CI [0.432, 0.517]; all frozen gates pass.

Table 6. Scene-level v6 confirmation. Intervals are dataset-stratified scene-bootstrap 95% intervals over 20 scenes; pooled case totals serve as descriptive evidence, with inference at scene level.

Metric	Result
Scenes with non-zero opacity LOCAL	20 / 20
Scenes with opacity tolerance crossover	20 / 20
Opacity non-zero-local rate	0.477 [0.432, 0.517]
Translation non-zero-local rate	0.000 [0.000, 0.000]
Same-opacity legacy-envelope rescue rate	0.531 [0.512, 0.549]
Largest scene maximum of residual / epsilon	0.0812
Median scene residual effectivity	1168.2

Gaussian editing, and continual mapping methods determine how scene state is represented or changed. CBRC sits after an edit has been specified and asks a different systems question: which derived blocks can remain stale without exceeding an output contract? Exact graph closure handles structural correctness, analytic bounds handle supported finite influence, the work model handles whether reuse is worthwhile, and the independent full-after path checks the resulting certificate. The broad campaign exercises implementation fidelity; the residual-sensitive sequence separately tests whether bounded non-zero output error can be admitted rather than forcing exact equality.

8.2 Limitations

Correctness and certificate tightness remain separate. The broad campaign is exact at the audited 8-bit raster, while v5 and v6 demonstrate certified non-zero residuals. That closes the earlier evidence gap while leaving certificate tightness unresolved. Accepted v6 cases remain comfortably inside the tolerance boundary and residual effectivity remains high (Table 6), showing substantial safety margin alongside substantial conservatism. The supported claim is repeatable residual-sensitive locality under the tested opacity-only assumptions; tighter display-space error estimation remains open.

Calibrated work is distinct from runtime. The headline work result is a frozen four-domain accounting ratio. It captures heterogeneous work under a versioned calibration while omitting allocator behavior, synchronization, cache locality,

command submission, and pipeline overlap. Paired end-to-end timing therefore remains a separate matched-runtime study.

Representation coverage remains output-side. The broad campaign is materially wider than the earlier four-scene experiment: it contains 72 dataset-native geometry worlds and 13 trained 3DGS scenes. Even so, the measured revision path concentrates on Gaussian inspection/update, GPU publication, and temporal output invalidation. The wider observation/TSDf/mesh/texture/material ledger is implemented and regression-tested; an equally broad performance campaign for that wider ledger remains future work.

Dataset-normalized worlds measure revision fidelity. For 3RScan, ARKitScenes, and Bonn, the benchmark evaluates deterministic seeded Gaussian representations derived from public captured-world assets. The trained GraphDECO subset evaluates pinned trained 3DGS. Selected-to-FULL equality therefore measures revision fidelity within each evaluated representation. Photorealistic reconstruction quality against held-out photographs lies outside this benchmark’s scope.

Some mechanisms are weakly identifiable on the real matrix. Several ablations tie the full method on aggregate work and certification. We keep those ties visible. The mechanism-isolation tests establish that those components can become consequential under controlled stress, while the broad matrix shows limited influence from them on aggregate real-scene outcomes.

The planner is deliberately conservative. Uniform-scale and adversarial cases trigger many full rebuilds. Those fallbacks are part of the intended behavior: when exterior influence, temporal stability, or work crossover cannot support a local result, the system prefers a full revision. The selected cones carry no global optimality guarantee.

8.3 When the method is useful

CBRC is useful when the persistent representation exposes stable ownership, publication, and output-influence structure. If every edit globally changes visibility, topology, or temporal validity, the revision cone expands to FULL and the method provides no computational advantage. Likewise, an overly conservative analytic bound can erase locality even when the

actual residual is small; the v4 diagnostic demonstrates exactly that failure. The v5/v6 opacity experiment shows the complementary case: making the certificate sensitive to the actual alpha change restores local decisions without changing the physical edit or the translation control.

This distinction is important operationally. A FULL result is not hidden as an implementation failure; it states that the current dependency and error model cannot justify leaving enough state untouched. Conversely, a LOCAL result is meaningful only for the declared protected output and tolerance. Improving the method therefore means either exposing more exact dependency structure, deriving tighter conservative bounds, or reducing the actual work of the selected repair—not simply forcing more cases to be local.

9 REPRODUCIBILITY AND ARTIFACT SCOPE

The base completion manifest binds the 1,275 campaign rows, 85 prepared worlds, dataset import manifests, frozen case matrix, work calibration, visual audit, cross-dataset statistics, tool fingerprints, and evidence Git SHA. The residual-sensitive line keeps v4 and v5 immutable and freezes v6 independently at 20 scenes and 3,840 cases. Its retained evidence includes the frozen campaign and provenance, canonical rows and baselines, scene-level metrics, the generic and v6 audits, the manuscript claim packet, case-decision trace, and deterministic reviewer-figure index. Cases resume only under matching deterministic fingerprints; bulky per-case render material is reproducible and may be compacted after the canonical row, baseline, replay manifest, and completion marker are retained.

Public datasets remain upstream assets, and the repository excludes their redistribution. The repository contains the dense reference certifier, native planner and Gaussian oracle, trained and seeded representation paths, baseline and ablation suites, deterministic statistics and visual-audit generators, residual-sensitive certificate tests, regression tests, and storage-safe campaign runners. For double-blind review, identifiable repository and author metadata should be replaced with an anonymized frozen supplement that follows the target venue’s policy.

10 CONCLUSION

Persistent 3D reconstructions need an update rule between two extremes: rebuilding the entire scene after every change and trusting a geometrically local edit without accounting for the derived state it leaves behind. CBRC provides that rule. It treats the recomputed state as a revision cone, closes structural dependencies exactly, bounds supported influence outside the cone at the protected output, compares regional work with a full baseline, and returns FULL whenever those conditions do not justify reuse.

In the 85-scene broad campaign, the algorithm selects 935 local repairs and 340 full fallbacks with no observed certificate violation under independent full-after replay; median

calibrated selected work is 0.474 of the full baseline. The separate residual-sensitive study shows why certificate design matters: the original magnitude-insensitive bound rejects every non-zero residual case, while the opacity-delta refinement restores non-zero local decisions in all 20 confirmatory scenes and leaves the unchanged translation control at zero. The remaining challenge is tighter conservative output bounds and matched end-to-end runtime evaluation. The bounded claim is therefore straightforward: persistent graphics state can be recomputed selectively when the implementation can justify both the residual left outside the update and the work saved by retaining it.

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Generative AI use disclosure. OpenAI ChatGPT was used for literature-search support, software and scripting assistance, debugging, experiment planning, visualization and manuscript tooling, and drafting or language editing. Reported measurements and frozen evidence came from project software and experimental artifacts. The author is responsible for the code, citations, analysis, claims, conclusions, and final text; no generative AI system is listed as an author.

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